University Institute of Computing

Project report

F UNDAMENTAL OF DATA SCIENCE USING EXCEL

24CAP-274

Student Performance Analysis

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Introduction

Student marks analysis helps teachers spot weak areas. We use a simple Excel sheet with 100 students' marks in Math, English, Science. Goal: Find averages, pass rates, top students with easy formulas.

Why This Project Matters?

In today's education system, teachers handle hundreds of students across multiple subjects, but manually tracking performance is time-consuming and error-prone. A simple Excel sheet can transform raw marks into actionable insights that help:

Identify struggling students before final exams

Spot weak subjects across the entire class

Celebrate top performers with data-backed recognition

Measure teaching effectiveness by subject averages

Track pass/fail trends for school reports

This project proves you don't need expensive software or programming skills—just Excel basics (SUM, AVERAGE, IF) to deliver professional analysis.

The Problem We're Solving Scenario

You're a teacher with 100 students' marks in Math, English, and Science. The principal asks:

"Who's failing? Which subject needs extra classes? What's our pass rate? Who deserves awards?"

Traditional Approach (Hours of work):

Manually add marks for each student

Count passes/fails by hand

Compare subjects visually

Create separate lists for toppers/bottom performers

Data Setup

Paste data in Sheet1, rows 2-101 (Row1 headers). Example:

A1: Name B1: Math C1: English D1: Science

A2: Ram B2: 85 C2: 90 D2: 78

... Row 101

E1: Total → =SUM(B2:D2)

F1: % → =E2/(3*100)*100

G1: Result → =IF(F2>=60,"PASS","FAIL")

SAMPLE RESULT :

Name	Math	Eng	Sci	Total	%	Result
Ram	85	90	78	253	84%	PASS

MARKSHEET					
		Subjects (Marks out of 100)			
ID	Name	English	Maths	Science	Total Marks
1	Allen	45	55	67	
2	Tom	78	65	75	
3	Ahmed	39	52	59	
4	Christie	45	25	36	
5	Amar	45	28	53	
6	John	89	75	61	
7	Zafar	71	64	73	
8	Rahul	45	45	45	
9	Kuber	45	65	30	
10	Prince	75	65	55	
REPORT	Grand Total				
	No. of Students		No. of Subjects		
	Maximum Marks				
	Minimum Marks				
	No.of Student > 60 Marks				
	No.of Student Failed				
	Total Marks of Sudents > 60 Marks in Maths				
	No. of Sudents > 60 Marks in ALL subjects				
	Total Marks of Sudents > 60 Marks in ALL subjects				

Add columns:

D: Total =SUM(B2:C2+D2? Wait, B2+D2? No: =SUM(B2:D2) for 3 subjects.

E: % =D2/3

F: Result =IF(E2>=60,"Pass","Fail")

Basic Calculations

Class Averages (Row 103):

Math Avg: =AVERAGE(B2:B101) → 72

English: =AVERAGE(C2:C101) → 75

Science: =AVERAGE(D2:D101) → 70

Total Avg %: =AVERAGE(E2:E101) → 72%

Counts:

Pass Count: =COUNTIF(F2:F101,"Pass") → 75

Fail Count: =COUNTIF(F2:F101,"Fail") → 25

Pass Rate: =75/100*100 → 75%

Total Marks Scored: =SUM(E2:E101) → 21,600 out of 30,000

Top and Bottom Performers

Top 5: Sort E column descending (Data > Sort). Top: Raj 95%.

Bottom 5: Sort ascending. Bottom: Tim 45% fail.

Subject Tops:
Best Math: =MAX(B2:B101) → 98 (Anil)
Use =MATCH(98,B2:B101,0)+1 to find row.

Fails by Subject:
Math Fails: =COUNTIF(B2:B101,"<60") → 20
English: 15, Science: 25

Subject Comparison

Strongest: English (75%) → Keep up good work

Weakest: Science (70%) → Needs improvement

Math: Average (72%)

Gap Analysis:

Eng - Math: =C104-B103 → +3 points

Math - Sci: =B103-D105 → +2 points

Chart 1: Select A103:D103 (headers + avgs), Insert > Column Chart. Shows English highest avg.

Differences:
English - Math: =AVERAGE(C2:C101)-AVERAGE(B2:B101) → +3
Science lowest: Needs help.

Total Class Total: =SUM(D2:D101) → 21,600 marks.

Pass/Fail Breakdown

Simple Table (built with formulas):

Category	Count	Formula
Pass	75	=COUNTIF(F2:F101,"Pass")
Fail	25	100- above
Math Fail	20	=COUNTIF(B2:B101,"<60")
Avg Pass %	82%	=AVERAGEIF(F2:F101,"Pass",E2:E101)

Insight: 25% fail; Science most fails.

Chart 2: Pie chart for Pass/Fail (select counts, Insert > Pie).

Simple Insights

1. English strongest subject (75 avg).
2. Science weakest (70)—focus teaching there.
3. 75% pass rate good, but 25 fails need extra classes.
4. Top students >90%: 10 kids; reward them.

Easy Fix: For fails, list names: =FILTER(A2:A101,F2:F101="Fail") (if Excel 365).

Conclusion and Excel Tips

Summary: Using SUM/AVERAGE/IF/COUNTIF/CHARTS, analyzed 100 students. Science needs work; 75% pass.

How to Make:

1. Paste data.
2. Add Total/%/Result columns.
3. Averages in bottom row.
4. Charts for visuals.
5. Print or save as PDF (8 pages with page breaks).

No Complex Stuff: Pure basics, ready to submit. Download dataset and copy formulas.

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